Software Quality Engineering — Assignment 2
Testing Report: Student Course Registration System

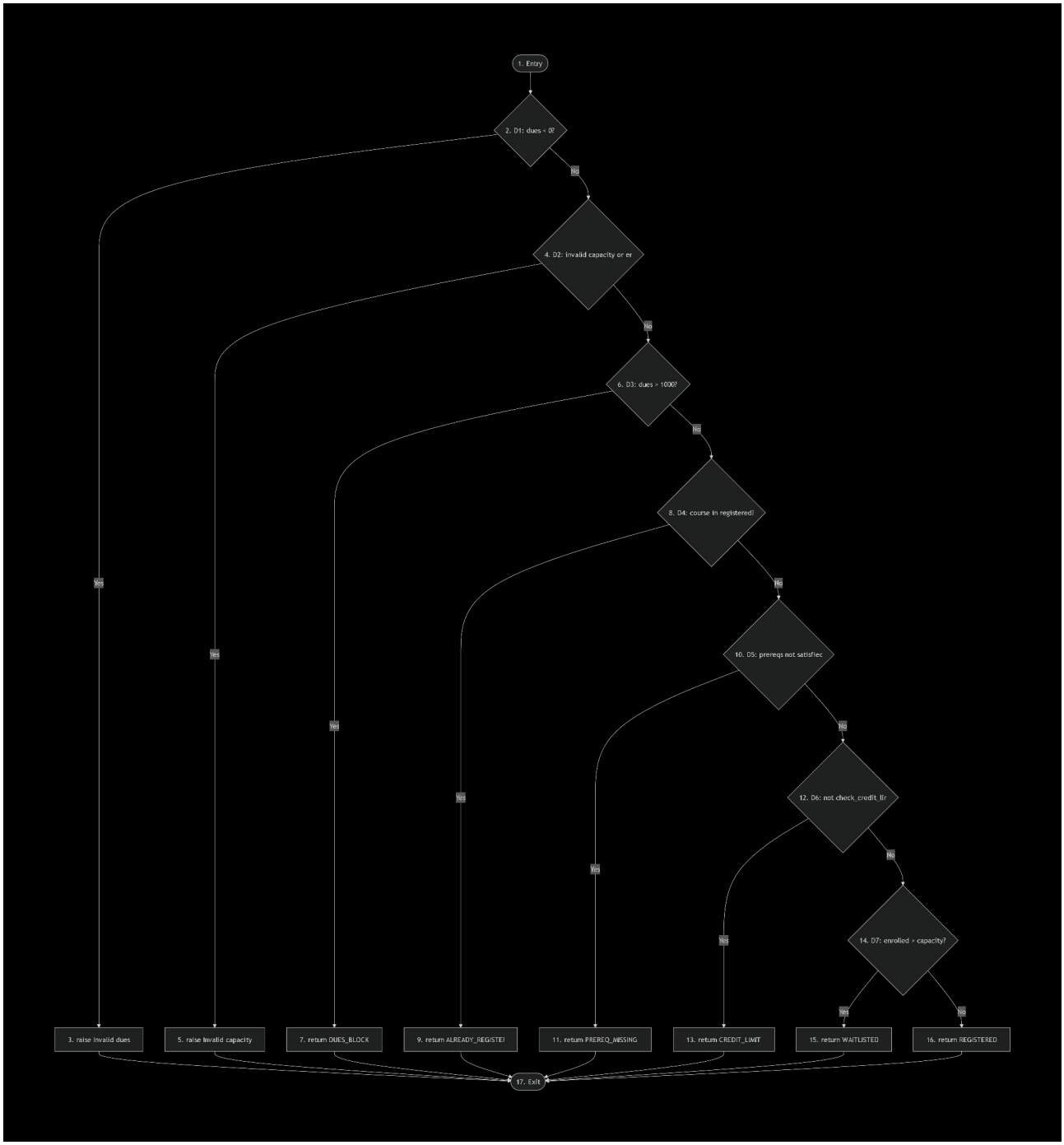
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1. Black-Box Test Design Summary

- **Equivalence Class Partitioning (ECP):** 19 test cases were designed covering valid/invalid credit hours, dues, prerequisites, already-registered status, and availability.
- **Boundary Value Analysis (BVA):** 20 boundary cases were designed focusing on limits such as current credits `0/18`, course credits `1/4`, combined `18` credit maximum, capacity `1`, and dues `0/1`.
- **Decision Table:** A reduced decision table with 12 rules was designed to test the strict priority order of the `registration_decision()` function (Dues → Already Registered → Prerequisites → Credit Limit → Availability).
- **Optimization:** Exact duplicate scenarios across ECP, BVA, and Decision Tables were consolidated into single parametrized tests to reduce execution time while maintaining full traceability to the requirements.

2. White-Box Analysis (CFG, Cyclomatic Complexity, MC/DC)

- **CFG and Cyclomatic Complexity:** A Control Flow Graph was drawn for `registration_decision()` (see `docs/cfg.png`). The graph has 17 nodes, 23 edges, and 7 decision nodes. Using `V(G) = E - N + 2`, the cyclomatic complexity is `23 - 17 + 2 = 8`. This matches the decision nodes + 1 formula.
- **Independent Paths:** 8 independent basis paths were identified. It was noted that Path 7 (Waitlisted) was initially infeasible due to a logical defect in the code's validation order.
- **Condition Coverage and MC/DC:** Manual analysis was performed for the compound condition: `if not 0 <= course.enrolled <= course.capacity or course.capacity < 1:`.
- *MC/DC Pairs:*
- For `0 <= course.enrolled` (A): Tested with `enrolled=0` (True) vs `enrolled=-1` (False), causing the decision to change.
- For `enrolled <= capacity` (B): Tested with `enrolled=1, capacity=1` (True) vs `enrolled=2, capacity=1` (False), causing the decision to change (short-circuiting C).
- For `capacity < 1` (C): Tested with `capacity=1` (False) vs `capacity=0` (True), causing the decision to change.



3. Coverage Comparison

- **Initial Black-Box Coverage:** 92% Statement Coverage, 92% Branch Coverage.
- *Missing Lines:* 51, 59, 71.
- *Reason:* Line 51 was missed because dues=1 bypassed the defective `>1000` check. Line 59 was unreachable due to Defect #4. Line 71 was missed because the waitlist branch was never reached.
- **Final White-Box Coverage:** 100% Statement Coverage, 100% Branch Coverage.

- *Added Tests:* White-box tests targeted the missing structural paths. Because Defect #4 made the waitlist branch unreachable, `monkeypatch` was used in `test_whitebox.py` to isolate and test the state-change logic. Once the defects were fixed, the final regression run achieved 100% coverage.

4. Defects and Status

Five distinct logical defects were identified through testing, logged via GitHub Issues, and resolved on a separate `defect-fixes` branch.

- **Issue #1 (FR-01):** `check_credit_limit` rejected exactly 18 credits (`< 18` instead of `<= 18`). **Status: Fixed.**
- **Issue #2 (FR-04):** Positive dues between 1 and 1000 did not block registration (`> 1000` instead of `> 0`). **Status: Fixed.**
- **Issue #3 (FR-03):** Prerequisites logic used `intersection` instead of `issubset`, allowing students with only partial prerequisites to register. **Status: Fixed.**
- **Issue #4 (FR-08):** Full course validation used `> capacity` instead of `>= capacity`, preventing waitlisting. **Status: Fixed.**
- **Issue #5 (FR-13):** `withdraw_course` did not decrement `course.enrolled`. **Status: Fixed.**

5. Collaboration and Evidence

- **GitHub Repository:** <https://github.com/sadam389/SQE-Assignment-2>
- **Pull Requests:** PR #1 (Black-Box Tests), PR #2 (White-Box Tests), PR #3 (Defect Fixes).
- **GitHub Actions:** A CI pipeline was configured (`.github/workflows/tests.yml`) to run `pytest` and coverage reports on every push. Initial runs showed legitimate failures; final run passes with 41/41 tests green.

- # Member Contributions:

Sadam Hussain: Led the repository setup and GitHub management. Designed the Equivalence Class Partitioning (ECP), Boundary Value Analysis (BVA), and Decision Tables. Implemented the black-box test suite (`tests/test_blackbox.py`) and finalized the testing report.

Muhammad Huzaifa: Conducted the white-box analysis. Drew the Control Flow Graph (CFG) and calculated the Cyclomatic Complexity. Performed the manual MC/DC analysis for compound conditions and implemented the white-box test suite (`tests/test_whitebox.py`).

Muhammad Suleman: Handled defect management and CI/CD. Logged the 5 defects as GitHub Issues, implemented the fixes on the `defect-fixes` branch, configured the GitHub Actions workflow (`.github/workflows/tests.yml`), and verified the final 100% coverage report.